

→ you can approach it ^{with} ~~from~~ elements from your set. 3.10.2025
Lecture 5

1) Accumulation point. → get as close as you want with other elem.

$A \subseteq \mathbb{R}$, $x_0 \in \bar{A}$ is acc. point if $(\forall) \forall \epsilon > 0, V \cap (A \setminus \{x_0\}) \neq \emptyset$.

$A' \rightarrow$ set of accumulation points.

2) Isolated point: $x_0 \in A$, if $x_0 \in A \setminus A'$

(negation of the acc. point def.)

Example: ① $A = \left\{ \frac{1}{m} \mid m \in \mathbb{N} \right\}$

$A' = ?$

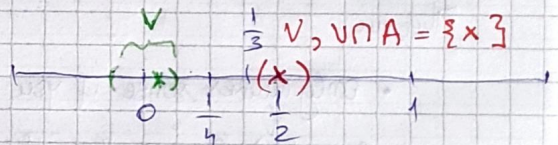
→ isolated point

what nr. always has some neighbours from our set? 0.

• $(\forall) x \in A$ is an isolated point

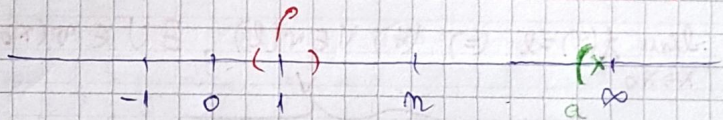
• 0 is an accumulation point

$(0 \in A')$



there's no one around it.

② $B = \mathbb{N}$



• $(\forall) m \in \mathbb{N}$ is an isolated point.

Take $V = (m-1, m+1) \cap \mathbb{N} = \{m\}$

↓
the only elem. in there
⇒ isolated.

• $\infty \in A' \Rightarrow$ accumulation point.

$cl(A) = A' \cup \{\text{isolated points of } A\}$.

Acc. point → can be approximated using other elems. of the set

Proposition: $A \subseteq \mathbb{R}$; then $x_0 \in A' \Leftrightarrow \exists x_m \in A \setminus \{x_0\}$ s.t.

$\lim_m x_m = x_0$.

Limit of a function: Def:

$A \subseteq \mathbb{R}$, $f: A \rightarrow \mathbb{R}$, $x_0 \in A'$ $\rightarrow$ acc. point.

$\lim_{x \rightarrow x_0} f(x) = l \in \overline{\mathbb{R}}$ if:

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$$

for any neighbourhood of the limit, you will find ...

$$(\forall) \forall \epsilon \in \mathcal{V}(l), \exists U \in \mathcal{V}(x_0) \text{ s.t. } f(x) \in U, (\forall) x \in U \cap (A \setminus \{x_0\})$$

Theorem: $A \subseteq \mathbb{R}$; $x_0 \in A'$. Then $\lim_{x \rightarrow x_0} f(x) = l \in \overline{\mathbb{R}}$ ($\Leftrightarrow$)

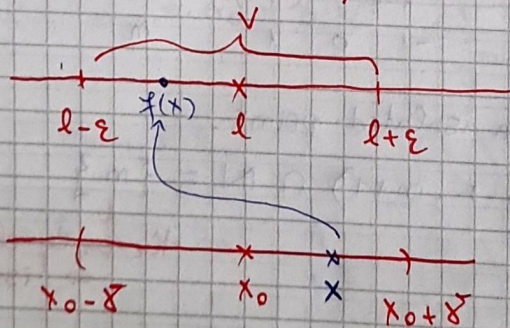
for any sequence (x_n) in $A \setminus \{x_0\}$ with $\lim_n x_n = x_0$, we have that $\lim_n f(x_n) = l$.

Theorem:

- only makes sense if you CAN approach x_0 from $\left\{ \begin{array}{l} \text{left} \\ \text{right} \end{array} \right.$

$$\lim_{x \rightarrow x_0} f(x) = l \Leftrightarrow \lim_{x \nearrow x_0} f(x) = \lim_{x \searrow x_0} f(x) = l$$

$$\lim_{x \rightarrow x_0} f(x) = l \Leftrightarrow (\forall) \forall \epsilon \in \mathcal{V}(l), \exists U \in \mathcal{V}(x_0) \text{ s.t. } f(x) \in U, (\forall) x \in U \cap (A \setminus \{x_0\})$$



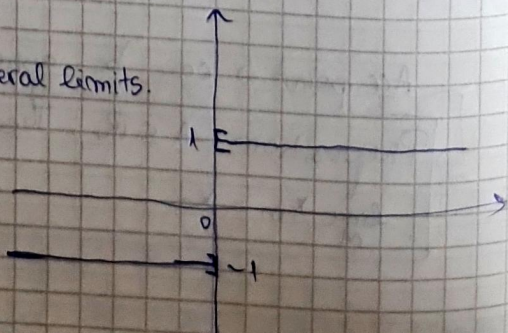
Translate: if you want $f(x)$ to be in the neighbourhood of the limit, you just have to be close enough to $x_0 \Rightarrow$ you have a limit.

Ex: a) $\text{sgn}(x) = \begin{cases} +1, & x \geq 0 \\ -1, & x < 0 \end{cases}$

$\Rightarrow$ lateral limits.

Does this function have a limit in 0?

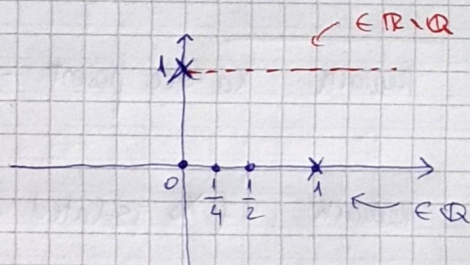
$\nexists \lim_{x \rightarrow 0} \text{sgn}(x)$



$$\lim_{x \nearrow 0} \operatorname{sgn}(x) = -1 \quad \nRightarrow \quad \nexists \lim_{x \rightarrow 0} f(x)$$

$$\lim_{x \searrow 0} \operatorname{sgn}(x) = 1$$

$$b) \quad f(x) = \begin{cases} 0, & x \in \mathbb{Q} \\ 1, & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$



Let $x_0 \in \mathbb{R}$. $\nexists \lim_{x \rightarrow x_0} f(x)$
(the theorem with the seq.)

because we can take $(x_m) \in \mathbb{Q}$, $x_m \rightarrow x_0$, $f(x_m) = 0$

$(\bar{x}_m) \in \mathbb{R} \setminus \mathbb{Q}$, $\bar{x}_m \rightarrow x_0$, $f(\bar{x}_m) = 1$

$$\left. \begin{aligned} \lim_{m \rightarrow \infty} f(x_m) &= \lim_{m \rightarrow \infty} f(\bar{x}_m) \\ \lim_{m \rightarrow \infty} x_m &= \lim_{m \rightarrow \infty} \bar{x}_m = x_0 \end{aligned} \right\} \Rightarrow \nexists \lim_{x \rightarrow x_0} f(x).$$

$$\lim_{m \rightarrow \infty} x_m = \lim_{m \rightarrow \infty} \bar{x}_m = x_0$$

$$c) \quad \sin \frac{1}{x}, \quad \nexists \lim_{x \rightarrow 0} \sin \left(\frac{1}{x} \right) \text{ because } x_m \rightarrow 0, f(x_m) = 1$$

$$y_m \rightarrow 0, f(y_m) = -1$$

$$x_m = \frac{1}{2m\pi + \frac{\pi}{2}} \xrightarrow{m \rightarrow \infty} 0, \quad \frac{1}{x_m} = 2m\pi + \frac{\pi}{2}$$

$$\sin \left(2m\pi + \frac{\pi}{2} \right) = \sin \left(\frac{\pi}{2} \right) = 1.$$

$$y_m = \frac{1}{2m\pi + \frac{3\pi}{2}} \xrightarrow{m \rightarrow \infty} 0, \quad \frac{1}{y_m} = 2m\pi + \frac{3\pi}{2}$$

$$\sin \left(2m\pi + \frac{3\pi}{2} \right) = \sin \left(\frac{3\pi}{2} \right) = -1.$$

$$\Rightarrow \nexists \lim_m \sin \frac{1}{x}$$

Continuous functions.

Def: $A \subseteq \mathbb{R}$, $f: A \rightarrow \mathbb{R}$, $x_0 \in A$; f is cont. at x_0 if

$(\forall) \epsilon \in \mathbb{N}(f(x_0))$, $\exists U \in \mathbb{N}(x_0)$ s.t. $f(x) \in U$, $(\forall) x \in U \cap A$.

Remark: x_0 acc. point $\Rightarrow f$ cont. at x_0 if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$

Remark: if x_0 isolated point of $A \Rightarrow f$ continuous at x_0 .

Theorem: $A \subseteq \mathbb{R}$, $f: A \rightarrow \mathbb{R}$, $x_0 \in A \cap A'$

1) f cont. at x_0

2) $(\forall) \epsilon > 0$, $\exists \delta > 0$ s.t. $|f(x) - f(x_0)| < \epsilon$, $(\forall) x \in A$, with $|x - x_0| < \delta$.

3) $\lim_{n \rightarrow \infty} x_n = x_0 \Rightarrow \lim_{n \rightarrow \infty} f(x_n) = f(x_0)$, $(\forall) x_n \in A$.

! Elementary functions are continuous (when they are well-defined)

Def: $f: A \rightarrow \mathbb{R}$

$$f(A) := \{f(x) \mid x \in A\}$$

f is bounded if $f(A)$ is a bounded set

$$\inf_{x \in \mathbb{R}} \quad \sup_{x \in \mathbb{R}}$$

Weierstrass Theorem:

$f: [a, b] \rightarrow \mathbb{R}$ cont.

f is bounded and it attains its bounds

$(\exists \min f(A), \exists \max f(A))$.

Theorem (Dorbox ~ Intermediate value property)

$f: [a, b] \rightarrow \mathbb{R}$ cont.

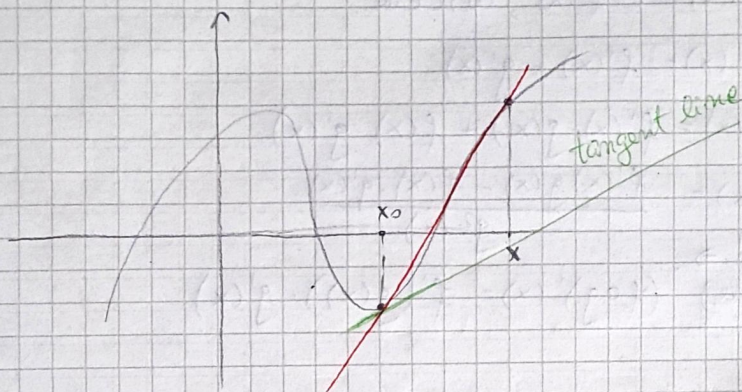
f has the i. v. property if $(\forall) y \in \mathbb{R}$ in between $f(a)$ and $f(b)$,

$\exists c \in (a, b)$ s.t. $f(c) = y$.

Derivatives

? $m_d = f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \in \mathbb{R} \rightarrow \mathbb{R} \cup \{-\infty, \infty\}$ the derivative of f at x_0 (are derivată în x_0)

if $f'(x_0) \in \mathbb{R}$ (finite) $\Rightarrow f$ differentiable at x_0 .
(derivabilă în x_0)



Equation of the tangent to the curve $y = f(x)$

$$f'(x_0)(x - x_0)$$

? $f(x) - f(x_0) = f'(x_0)(x - x_0)$

Graph has corners $\Rightarrow$ non-differentiable

has jumps $\Rightarrow$ discontinuous

differentiable $\Rightarrow$ continuous

continuous $\nRightarrow$ differentiable

$\hookrightarrow$ Proof: Let f diff. at x_0 : $f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \in \mathbb{R}$

I want to show that f cont. at x_0 : $\lim_{x \rightarrow x_0} f(x) = f(x_0)$

$$f(x) = f(x_0) + \overbrace{(x - x_0)}^{f'(x_0)} \cdot \frac{f(x) - f(x_0)}{x - x_0} \quad \Bigg\} \Rightarrow$$

Take the limit $\lim_{x \rightarrow x_0}$

$$\Rightarrow \lim_{x \rightarrow x_0} f(x) = f(x_0) + 0 \cdot f'(x_0) = f(x_0)$$

Brownian motion $\rightarrow$ they are continuous, but as you zoom in, the pattern repeats $\rightarrow$ non-differentiable.

Calculus rules:

- 1) $(c \cdot f)'(x) = c \cdot f'(x), (\forall) c \in \mathbb{R}$.
- 2) $(f+g)'(x) = f'(x) + g'(x)$
- 3) $(f \cdot g)'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$
- 4) $\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$
- 5) $u \cdot u' \rightarrow$
(Chain rule) $(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$

L'Hôpital:

$$\frac{0}{0}, \frac{\infty}{\infty}$$

$$\exists \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} \in \overline{\mathbb{R}} \Rightarrow \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)}$$

$f: A \rightarrow \mathbb{R}$ has a local extremum at $x_0 \in A$ if ... s.t. $f(x_0) \leq f(x)$
or

$\downarrow$
not about the entire domain,
but about the neighbourhood

$f(x_0) \geq f(x)$
maximum

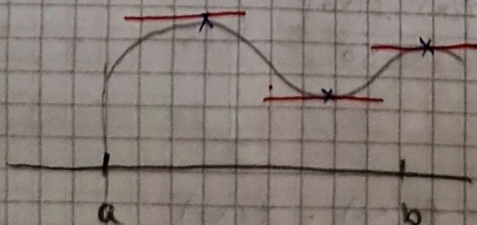
pt. de min vs. min

pt. de max vs. max

Fermat theorem:

$f: (a,b) \rightarrow \mathbb{R}, x_0 \in (a,b)$ a local extremum.

f is differentiable at x_0 if $f'(x_0) = 0$.



slope of the tangent : 0. (horizontal)

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Let x_0 be a local minimum $\Rightarrow$

$$\Downarrow \\ f(x) \geq f(x_0)$$

$$\lim_{x \nearrow x_0} \frac{\overbrace{f(x) - f(x_0)}^{\geq 0}}{\underbrace{x - x_0}_{> 0}} \geq 0$$

$$\lim_{x \searrow x_0} \frac{\underbrace{f(x) - f(x_0)}_{\leq 0}}{\underbrace{x - x_0}_{< 0}} \leq 0$$

$$\Rightarrow L_1 = L_2$$

$$L_1 \geq 0$$

$$L_2 \leq 0$$

$$\Downarrow$$

$$L_1 = L_2 = 0$$

$$\Downarrow$$

$$f'(x_0) = 0$$

$$f \text{ is differentiable} \Rightarrow L_2 = \lim_{x \nearrow x_0} \dots = \lim_{x \searrow x_0} \dots = L_1$$

Rolle Theorem

$$f: (a, b) \rightarrow \mathbb{R}$$

$$f(a) = f(b)$$

f cont. on $[a, b]$

f diff. on (a, b)

$$\Rightarrow \exists c \in (a, b) \text{ s.t. } f'(c) = 0$$

Lagrange (~ theorem de medie) Mean value

$$f: (a, b) \rightarrow \mathbb{R}$$

$$f(a) \neq f(b)$$

f cont. on $[a, b]$

f diff. on (a, b)

$$\Rightarrow \exists c \text{ s.t. } f'(c) = \frac{f(b) - f(a)}{b - a}$$

Theorem (monotony)

$$f: (a, b) \rightarrow \mathbb{R} \text{ diff on } (a, b)$$

$$\Rightarrow f \nearrow \Leftrightarrow f' \geq 0$$

$$\Rightarrow f \searrow \Leftrightarrow f' \leq 0$$